

Task 5 Output

Avgpool, k=5, svd

```
Label - weight pairs sorted in descending order of weights for each latent feature:

Latent feature: 1
(Label: schooner , Weight: -0.048563292359567844), (Label: ketch , Weight: -0.057495824822448946), (Label: tick , Weight: -0.05940119997536563), (Label: Leopard , Weight: -0.06030119997536563), (Label: crocodile , Weight: -0.06030119997536563)

Latent feature: 2
(Label: emu , Weight: 0.19469201064974476), (Label: kangaroo , Weight: 0.18741296011920958), (Label: crocodile , Weight: 0.17619454406833782), (Label: geronimo , Weight: 0.17619454406833782), (Label: crocodile , Weight: 0.17619454406833782)

Latent feature: 3
(Label: water_lilly , Weight: 0.20927346166202257), (Label: lotus , Weight: 0.1911292399888913), (Label: sunflower , Weight: 0.18739012642903008), (Label: dragonfly , Weight: 0.18739012642903008), (Label: dragonfly , Weight: 0.18739012642903008)

Latent feature: 4
(Label: ketch , Weight: 0.3317858019869923), (Label: schooner , Weight: 0.293142029658296), (Label: ferry , Weight: 0.2607186166948226), (Label: pyramid , Weight: 0.2607186166948226), (Label: pyramid , Weight: 0.2607186166948226)

Latent feature: 5
(Label: sunflower , Weight: 0.28435405383738976), (Label: lotus , Weight: 0.24035665609183773), (Label: water_lilly , Weight: 0.21537934011076906), (Label: cup , Weight: 0.21537934011076906), (Label: cup , Weight: 0.21537934011076906)
```

Avgpool, k=5, nnmf

```
Label - weight pairs sorted in descending order of weights for each latent feature:

Latent feature: 1
(Label: dragonfly , Weight: 2.511893160169458), (Label: water_lilly , Weight: 2.4919374338486744), (Label: sea_horse , Weight: 2.471467131407958), (Label: sea_horse , Weight: 2.471467131407958), (Label: sea_horse , Weight: 2.471467131407958)

Latent feature: 2
(Label: cellphone , Weight: 3.170479850078846), (Label: stapler , Weight: 3.032728885973181), (Label: camera , Weight: 2.5497270126303357), (Label: wrench , Weight: 2.5497270126303357), (Label: wrench , Weight: 2.5497270126303357)

Latent feature: 3
(Label: pyramid , Weight: 2.622876251917305), (Label: helicopter , Weight: 2.555470105247227), (Label: pagoda , Weight: 2.5161180861759305), (Label: ferry , Weight: 2.5161180861759305), (Label: ferry , Weight: 2.5161180861759305)

Latent feature: 4
(Label: menorah , Weight: 2.750119859623749), (Label: chandelier , Weight: 2.614437194336191), (Label: chair , Weight: 2.467789866899583), (Label: gramophone , Weight: 2.467789866899583), (Label: gramophone , Weight: 2.467789866899583)

Latent feature: 5
(Label: brontosaurus , Weight: 2.86508323233955177), (Label: emu , Weight: 2.793186421511357), (Label: kangaroo , Weight: 2.7366871344482067), (Label: okapi , Weight: 2.7366871344482067), (Label: okapi , Weight: 2.7366871344482067)
```

Avgpool, k=5, lda

```
Label - weight pairs sorted in descending order of weights for each latent feature:

Latent feature: 1
(Label: octopus , Weight: 0.9816723191006395), (Label: anchor , Weight: 0.9810641647652497), (Label: brontosaurus , Weight: 0.979579839402593), (Label: gramophone , Weight: 0.979579839402593), (Label: gramophone , Weight: 0.979579839402593)

Latent feature: 2
(Label: schooner , Weight: 0.5587885370015079), (Label: ketch , Weight: 0.219334339103052), (Label: tick , Weight: 0.025196258738759376), (Label: Leopards , Weight: 0.025196258738759376), (Label: Leopards , Weight: 0.025196258738759376)

Latent feature: 3
(Label: schooner , Weight: 0.05348237155640449), (Label: ketch , Weight: 0.02580797051798783), (Label: tick , Weight: 0.025198937002239657), (Label: Leopards , Weight: 0.025198937002239657), (Label: Leopards , Weight: 0.025198937002239657)

Latent feature: 4
(Label: schooner , Weight: 0.05348237155641454), (Label: ketch , Weight: 0.025807970517992093), (Label: tick , Weight: 0.025198937002246936), (Label: Leopards , Weight: 0.025198937002246936), (Label: Leopards , Weight: 0.025198937002246936)

Latent feature: 5
(Label: schooner , Weight: 0.05348237155644787), (Label: ketch , Weight: 0.025807970518003393), (Label: tick , Weight: 0.02519893700225592), (Label: Leopards , Weight: 0.02519893700225592), (Label: Leopards , Weight: 0.02519893700225592)
```

Avgpool, k=5, kmeans

Label - weight pairs sorted in descending order of weights for each latent feature:

```
Latent feature: 1
(Label: octopus , Weight: 17.961237641597233), (Label: schooner , Weight: 17.74589237682219), (Label: anchor , Weight: 17.254462765246465), (Label: gramoj

Latent feature: 2
(Label: octopus , Weight: 22.712731731843883), (Label: anchor , Weight: 21.46591889469127), (Label: brontosaurus , Weight: 20.158674950114325), (Label: gramoj

Latent feature: 3
(Label: schooner , Weight: 25.43802379223996), (Label: ketch , Weight: 22.854972814346155), (Label: tick , Weight: 22.38188604451937), (Label: Leopards , We

Latent feature: 4
(Label: schooner , Weight: 23.26406826973131), (Label: ketch , Weight: 20.939009594341357), (Label: grand_piano , Weight: 19.279253375038255), (Label: Leopai

Latent feature: 5
(Label: schooner , Weight: 20.343354629126868), (Label: ketch , Weight: 18.12084108036494), (Label: tick , Weight: 17.1851633243398), (Label: Leopards , We
```

resnet, k=5, svd

Label - weight pairs sorted in descending order of weights for each latent feature:

```
Latent feature: 1
(Label: car_side , Weight: -0.012922288759228419), (Label: ferry , Weight: -0.01716962854338399), (Label: ibis , Weight: -0.02665104036861118), (Label: joshua

Latent feature: 2
(Label: brontosaurus , Weight: 0.22450847372794557), (Label: sea_horse , Weight: 0.20854212811298972), (Label: stegosaurus , Weight: 0.2012541405528358), (Label

Latent feature: 3
(Label: water_lilly , Weight: 0.19342605944553704), (Label: lotus , Weight: 0.17900652919063426), (Label: lobster , Weight: 0.1607137944172012), (Label: butte

Latent feature: 4
(Label: pyramid , Weight: 0.2559938442405397), (Label: bonsai , Weight: 0.25303356194758286), (Label: chandelier , Weight: 0.24746884141219433), (Label

Latent feature: 5
(Label: cup , Weight: 0.20185810889235156), (Label: strawberry , Weight: 0.1566181164612972), (Label: cougar_face , Weight: 0.15276962455955076), (Label: Faces
```

resnet, k=5, nnmf

Label - weight pairs sorted in descending order of weights for each latent feature:

```
Latent feature: 1
(Label: crayfish , Weight: 4.223983411446315), (Label: crab , Weight: 4.034828170802018), (Label: sea_horse , Weight: 3.8989223861455913), (Label: scorp:

Latent feature: 2
(Label: stapler , Weight: 4.3709470647179), (Label: wrench , Weight: 4.297474749008891), (Label: scissors , Weight: 4.208068665830369), (Label: headphone , W

Latent feature: 3
(Label: pyramid , Weight: 4.001624643923868), (Label: pagoda , Weight: 3.6155747794299433), (Label: minaret , Weight: 3.4635229230980364), (Label: helicopter , l

Latent feature: 4
(Label: lotus , Weight: 3.385984598158191), (Label: sunflower , Weight: 3.3764495990288452), (Label: garfield , Weight: 3.247732300281304), (Label: water:

Latent feature: 5
(Label: brontosaurus , Weight: 4.1733888754234885), (Label: rhino , Weight: 3.8140388759085275), (Label: okapi , Weight: 3.6970182416032045), (Label: gerenuk , Wei
```

resnet, k=5, lda

```
Label - weight pairs sorted in descending order of weights for each latent feature:

Latent feature: 1
(Label: ferry , Weight: 0.679151691635175), (Label: car_side , Weight: 0.5371101755634627), (Label: ketch , Weight: 0.48074845628223256), (Label: helicopter , Weight: 0.479151691635175)

Latent feature: 2
(Label: car_side , Weight: 0.33078643982630324), (Label: wheelchair , Weight: 0.16157168496151217), (Label: Motorbikes , Weight: 0.10032102881152959), (Label: car_side , Weight: 0.33078643982630324)

Latent feature: 3
(Label: octopus , Weight: 0.9738642270888856), (Label: menorah , Weight: 0.968092062395256), (Label: dragonfly , Weight: 0.9671523899422728), (Label: garfield , Weight: 0.9671523899422728)

Latent feature: 4
(Label: car_side , Weight: 0.040790709344990676), (Label: ferry , Weight: 0.03634283743526946), (Label: ibis , Weight: 0.027754165469637562), (Label: ketch , Weight: 0.027754165469637562)

Latent feature: 5
(Label: emu , Weight: 0.7826936955999719), (Label: rhino , Weight: 0.6993935522795274), (Label: ibis , Weight: 0.6666885227840709), (Label: gerenuk , Weight: 0.6666885227840709)
```

resnet, k=5, kmeans

```
Label - weight pairs sorted in descending order of weights for each latent feature:

Latent feature: 1
(Label: octopus , Weight: 20.0171715593660127), (Label: menorah , Weight: 17.06915537484625), (Label: garfield , Weight: 16.781800846633114), (Label: dragonfly , Weight: 16.781800846633114)

Latent feature: 2
(Label: emu , Weight: 15.64015224658079), (Label: joshua_tree , Weight: 15.454988259617966), (Label: rhino , Weight: 15.219654794135142), (Label: ibis , Weight: 15.219654794135142)

Latent feature: 3
(Label: octopus , Weight: 16.976980203217014), (Label: menorah , Weight: 15.8444442621784449), (Label: metronome , Weight: 15.752509034694699), (Label: metronome , Weight: 15.752509034694699)

Latent feature: 4
(Label: octopus , Weight: 14.228401770697667), (Label: metronome , Weight: 13.359848331778705), (Label: joshua_tree , Weight: 13.274062778423783), (Label: joshua_tree , Weight: 13.274062778423783)

Latent feature: 5
(Label: helicopter , Weight: 15.100649044501841), (Label: car_side , Weight: 15.059383077446176), (Label: ferry , Weight: 14.792013061635785), (Label: joshua_tree , Weight: 14.792013061635785)
```